MAKE-UP EXAMINATIONS (2022) QUESTION PAPER - PART A

COURSE CODE : CSE3153
COURSE NAME : OPERATING SYSTEMS
SEMESTER : V
DATE OF EXAM : 26 /02 /2022
DURATION : 45 + 5 minutes

Instructions for Students:
(1) ANSWER ALL THE QUESTIONS.
(2) EACH QUESTION CARRIES 1 MARK.
(3) YOU ARE INSTRUCTED TO INFORM THE INVIGILATOR AFTER SUBMISSION OF THIS FORM IN THE CHAT SECTION.

1. **User interaction with the computer system is possible in _____**

multi-programming

multi-tasking

Both multi-programming and multi-tasking

none of these

2. **_____ chooses a job if there are several jobs waiting to be executed at the same time in the main memory**

CPU scheduling

job scheduling

swapping

context switch

3. **The Intel 64 family of CPUs supports_____ privilege levels**

One

two

three

four

4. **Timer is used to _____**

control the operating system operations

to connect to the Internet

to control the user program execution on hardware

to control the kernel mode executions

5. **_____ is the fastest storage in computer system**

main memory

cache

register

magnetic disk

6. Choose the INCORRECT operating system operation from the following

recovery from failure

file-system manipulation

communications

error detection

7. _____ are the one who manage the computer

power user

system administrator

programmer

developer

8. Dirty bit of a page in a page table

a. Allows only read on a page
b. Helps to maintain LRU information
c. Helps to avoid unnecessary writes on paging device
d. Helps to avoid thrashing

9. Which of the following statement is false in case of virtual memory?
    a. It helps to translate logical address to physical address
    b. It allows program larger than main memory
    c. Increases degree of multiprogramming
    d. Reduces context-switch overhead
10. A system with byte-addressable memory, 32-bit logical addresses, 4KB page size and page table entries of 4 bytes each. The size of the page table in the system in MB is
    a. 2MB
    b. 4MB
    c. 6MB
    d. 8MB
11. A 1000 KB memory is managed using variable partitions allocation. It currently has two partitions of sizes 200KB and 260KB respectively. The smallest allocation request in KB that could be denied is for
    a. 541
    b. 150
    c. 179
    d. 230
12. LOOK scheduling is an extension of
    a. SSTF
    b. SCAN
    c. FCFS
    d. Its Own
13. A paging scheme uses a TLB. A TLB access takes 20ns and main memory access takes 60ns. What is the effective access time (in ns) if the TLB hit ratio is 90%?
    a. 90
    b. 50
    c. 100
    d. 80
14. Limit and base registers of a segment include 400 and 4300 respectively. The reference to a byte 530 of this segment is
    a. Legal reference
    b. Illegal reference
    c. Both legal and illegal
    d. Neither legal nor illegal
15. Which of the page replacement algorithm requires future knowledge of the string?
    a. Optimal
    b. FIFO
    c. LRU
    d. None of the above

16. Which of the following statements are true for the FCFS Scheduling algorithm?

i)FCFS uses FIFO queue for ordering ii) FCFS uses STACK structure for ordering iii) Process average waiting time is quite long iv) Optimal scheduling algorithm.

A) i, ii, iv
B) iii, iv
C) i, iii
D) i, ii, iii, iv

17. The multilevel feedback queue scheduler is defined by the following parameters.

i) The method used to determine when to block the process.

ii)The method used to determine when to demote a process to a lower-priority queue

iii)The method used to determine which queue a process will enter when that process needs service.

iv) The method used to determine when to terminate the process.

A)i, iv

B**) ii, iii**

C) i, ii, iii, iv

D) i, ii, iii

18. Choose the right option for wait for graph.

i)used when single instance of the resource.

ii)used when multiple instances of resources.

iii) used to detect cycle in a graph.

iv)used to avoid deadlock in operating systems.

A) ii, iii B) i, ii C) iii, iv D) ii, iii, iv

19. A preemptive kernel is not used in_____.

i)Real time processes

ii) Distributed processes

iii) cooperating process

iv) Windows Operating systems

20 . consider a system consisting of two processes, P0 and P1, each accessing two semaphores, S and Q, set to the value 1:

P0	P1
wait(S);	wait(Q);
wait(Q);	wait(S);
. . . . .	
signal(S);	signal(Q);
signal(Q);	signal(S);

After the execution of these statements system will be in_____.

A) deadlock state

B) deadlock state depends on the order of execution

C) never enter into deadlock

D) deadlock free state

21 An edge from process P_2 to P_1 in a wait for graph indicates that _____

A) P_1 is waiting for P_2 to release a resource that P_2 needs

b) P_2 is waiting for P_1 to release a resource that P_2 needs

c) P_1 is waiting for P_2 to leave the system

d) P_2 is waiting for P_1 to leave the system

22. A computer system has 12 tape drives, with 'n' processes competing for them. Each process may need 4 tape drives. The maximum value of 'n' for which the system is guaranteed to be deadlock free is?

A. 3

B. 4

C. 5

D. 2

23. Consider the following set of processes, arriving at the given times and having the following CPU burst time and priorities (Smaller number is having higher priority):

Process	Arrival Time(ms)	Burst Time (ms)	Priority
A	0	8	3
B	3	4	1
C	5	7	4
D	8	3	2

What is the turnaround time of **process C** using **Round Robin** scheduling algorithms (quantum time 3 ms).
A. 15
B. 13
C. 17

D. 11

24. _____ information is stored on the per-process table so the OS can allow or deny subsequent I/O requests.

a)user

b)BIOS

c)IPC

d)access rights

25 virtual Memory implemented through

a) Paging
b) Demand paging
c) Segmentation
d) Thrashing

26. "invalid bit" in a page table of demand paging indicates

a) Page is not legal

b) frame is illegal

c) page is in physical memory not in disk

d)frame is in smaller than page

27 The lowest cylinder movements in _____ disk scheduling algorithm

a) FCFS
b) SCAN

c) LOOK

d) SSTF

28. In which implementation method of Access Matrix uses ordered triplets < domain, oject, right-sets>

a) Access list for ojejcts

b) **Global Table**

c) Capaility List for domains

d) Lock-key mechanism

29. _____ begins at the root and follows a path down to the sspecified file, giving the directory names on the path.

a) class path

b) absolute path

c) Name

d) home

30 . Thhe location within the file structure where the file system is to e attached is called

a)restore point

b)mount point

c)root

d)directpath

Reg. No.										

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V SEMESTER B.TECH. (COMPUTER SCIENCE AND ENGINEERING)

MAKE-UP EXAMINATIONS 2022

SUBJECT: OPERATING SYSTEMS [CSE 3153]

REVISED CREDIT SYSTEM

(26 /02 /2022)

Time: 9:20am-10:35 am MAX. MARKS: 20

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitably assumed.

1A. With a neat diagram, illustrate how operating system distinguishes between the execution of operating-system code and user-defined code? **5M**

1B. Suppose that a disk drive has 200 cylinders, numbered from 0 to 199. The drive is currently serving a request at cylinder 60 and the queue of pending requests is: 85, 158, 38, 138, 34, 70, 64, and 13. Starting from current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all pending requests for each of the following disk-scheduling algorithms? **3M**

a) FCFS
b) SSTF
c) LOOK

1C. Illustrate the two-level directory structure with a neat figure **2M**

2A. The following processes are being scheduled using a FCFS, non-preemptive SJF, and Round-Robin (Time quantum- 3ms) scheduling algorithms. **5M**

Processes	Arrival Time	Burst Time
P1	2	4
P2	1	7
P3	4	3
P4	0	5
P5	3	2

a) Show the scheduling order of the processes using a Gantt chart.

b) What is the turnaround time for each process?

c) What is the response time for each process?

d) What is the waiting time for each process?

2B. A system has 12 resources of A type, 14 resources of B type and 15 resources of C type. Apply Banker's algorithm and check whether the system is safe using Bankers algorithm **3M**

Processes	MAX			Allocation			Available		
	A	B	C	A	B	C	A	B	C
P1	8	7	5	2	3	4	?	?	?
P2	7	4	4	4	2	3			
P3	2	5	5	1	4	4			
P4	5	5	5	5	5	4			

2C. There are 5 resources R1 with 2 instances, R2 with single instance, R3 with 2 instances, R4 with single instance and R5 with single instance **2M**

Process P1 holding one instance of R1 and an instance of R2 and requesting one instance of R5

Process P2 holding one instance of R1 and an instance of R4.

Process P3 is holding one instance of R3 and requesting one instance of R2.

Process P4 is holding one instance of R3 and requesting for one instance of R4 and requesting one instance of resource R5

Process P5 is requesting for one instance of R3 and holding one instance of R5.

Process P6 is requesting for one instance of R4.

Draw the resource allocation graph and identify whether deadlock exists? If yes, how many processes are involved in the deadlock? If No, give reason.Justify your answer depending on Yes/No.

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